

```

<!DOCTYPE html>
<html lang="zh-TW">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>iPAS AI 應用規劃師 - 終極備考系統</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <style>
    .perspective-1000 { perspective: 1000px; }
    .transform-style-3d { transform-style: preserve-3d; }
    .backface-hidden { backface-visibility: hidden; }
    .rotate-y-180 { transform: rotateY(180deg); }
  </style>
</head>
<body class="bg-slate-100 text-slate-800 min-h-screen pb-28">

  <!-- ===== 1. 安全通行驗證門禁 ===== -->
  <div id="authGate" class="fixed inset-0 bg-slate-900/95 backdrop-blur-md z-50 flex
items-center justify-center p-4">
    <div class="bg-white rounded-2xl max-w-sm w-full p-6 shadow-2xl border border-slate-200">
      <div class="flex items-center justify-center w-12 h-12 rounded-full bg-indigo-50
text-indigo-600 mx-auto mb-4 text-2xl">

      </div>
      <h2 class="text-base font-bold text-center text-slate-900 mb-1">私人備考系統</h2>
      <p class="text-xs text-center text-slate-500 mb-5">請輸入通行密碼與安全驗證碼以解鎖內容
</p>

      <div class="mb-3">
        <label class="block text-[11px] font-semibold text-slate-700 mb-1">通行密碼</label>
        <input type="password" id="inputPassword" placeholder="請輸入密碼..."
onkeydown="if(event.key==='Enter') handleUnlock()" class="w-full text-xs border
border-slate-300 rounded-lg px-3 py-2.5 focus:outline-none focus:ring-2 focus:ring-indigo-500"
/>
      </div>

      <div class="mb-4">
        <label class="block text-[11px] font-semibold text-slate-700 mb-1">安全驗證碼</label>
        <div class="flex items-center gap-2">
          <input type="text" id="inputCaptcha" placeholder="輸入 4 碼" maxlength="4"
onkeydown="if(event.key==='Enter') handleUnlock()" class="flex-1 text-xs border
border-slate-300 rounded-lg px-3 py-2 uppercase tracking-wider focus:outline-none focus:ring-2
focus:ring-indigo-500" />

```

```
<div id="captchaDisplay" onclick="refreshCaptcha()" title="點擊重新產生驗證碼"
class="cursor-pointer select-none bg-slate-100 border border-slate-300 rounded-lg px-3 py-1.5
font-mono text-sm font-bold tracking-widest text-indigo-700 line-through decoration-slate-400">
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</div>
```

```
<button type="button" onclick="refreshCaptcha()" class="text-slate-500
hover:text-slate-800 p-1 text-sm font-bold">
```



```
</button>
```

```
</div>
```

```
</div>
```

```
<div id="authErrorMsg" class="hidden text-[11px] text-rose-600 font-medium mb-3 p-2
bg-rose-50 rounded-lg border border-rose-200"></div>
```

```
<button id="btnUnlock" onclick="handleUnlock()" class="w-full bg-indigo-600
hover:bg-indigo-700 active:bg-indigo-800 text-white text-xs font-semibold py-2.5 rounded-lg
transition shadow-sm cursor-pointer">
```

```
  驗證並進入系統
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```
</button>
```

```
</div>
```

```
</div>
```

```
<!-- ===== 2. 主應用程式 ===== -->
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```
<div id="mainApp" class="hidden">
```

```
<!-- 頂部導覽列 -->
```

```
<header class="bg-white border-b border-slate-200 sticky top-0 z-30 shadow-xs">
```

```
<div class="max-w-5xl mx-auto px-4 py-2.5 flex items-center justify-between">
```

```
<div class="flex items-center space-x-2">
```

```
<span class="text-2xl">🧠</span>
```

```
<div>
```

```
<h1 class="text-sm font-bold text-slate-900 leading-tight">iPAS AI 終極備考系統</h1>
```

```
<span id="quizCountSummary" class="text-[10px] text-slate-500">載入中...</span>
```

```
</div>
```

```
</div>
```

```
<!-- 模組切換分頁導覽 -->
```

```
<nav class="flex items-center bg-slate-100 p-1 rounded-xl gap-1 text-xs font-medium
text-slate-600">
```

```
<button onclick="switchTab('quiz')" id="tabBtn-quiz" class="tab-btn active bg-white
text-indigo-600 px-3 py-1.5 rounded-lg shadow-xs transition">
```



```
  全真考場
```

```
</button>
```

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        <button onclick="switchTab('retry')" id="tabBtn-retry" class="tab-btn px-3 py-1.5
rounded-lg hover:text-slate-900 transition flex items-center gap-1">
            🔄 錯題盲測 <span id="navWrongCount" class="bg-rose-100 text-rose-600 text-[10px]
px-1.5 py-0.2 rounded-full font-bold">0</span>
        </button>
        <button onclick="switchTab('concepts')" id="tabBtn-concepts" class="tab-btn px-3 py-1.5
rounded-lg hover:text-slate-900 transition">
            📖 核心概念庫
        </button>
        <button onclick="switchTab('flashcards')" id="tabBtn-flashcards" class="tab-btn px-3
py-1.5 rounded-lg hover:text-slate-900 transition">
            📇 記憶翻牌卡
        </button>
        <button onclick="switchTab('analytics')" id="tabBtn-analytics" class="tab-btn px-3 py-1.5
rounded-lg hover:text-slate-900 transition">
            📊 做題分析
        </button>
    </nav>

```

```

    <div class="flex items-center space-x-2">
        <button onclick="openKeySettings()" class="text-xs bg-slate-100 hover:bg-slate-200
text-slate-700 px-2.5 py-1.5 rounded-lg flex items-center gap-1 border border-slate-300">
            🔑 <span id="keyStatusText">AI Key</span>
        </button>
        <button onclick="lockApp()" class="text-xs bg-slate-100 hover:bg-rose-50
hover:text-rose-600 text-slate-600 px-2.5 py-1.5 rounded-lg border border-slate-300 transition"
title="立即鎖定">
            🔒
        </button>
    </div>
</div>
</header>

```

```

<div class="max-w-5xl mx-auto px-4 mt-5">

```

```

    <!-- ===== TAB 1: 全真考場(整卷作答 + 交卷按鈕 + 紅綠對照線)
===== -->

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```

    <section id="tabContent-quiz" class="space-y-4">
        <!-- 考場功能選單 -->
        <div class="bg-indigo-900 text-white p-3.5 rounded-2xl shadow-sm flex flex-wrap
items-center justify-between gap-2 text-xs">
            <div class="flex flex-wrap items-center gap-2">
                <button onclick="loadOfficial790()" class="bg-emerald-600 hover:bg-emerald-700

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text-white font-medium px-3 py-1.5 rounded-lg transition">

 載入官方 790 題庫抽卷

</button>

<button onclick="startMockExam(50)" class="bg-indigo-700 hover:bg-indigo-600

text-white px-3 py-1.5 rounded-lg transition">

 75分鐘模考 (50題)

</button>

<button onclick="restartExam()" class="bg-indigo-800 hover:bg-indigo-700 text-white

px-3 py-1.5 rounded-lg transition">

 重新抽題測驗

</button>

</div>

<div id="examTimerContainer" class="hidden items-center bg-indigo-950 px-3 py-1 rounded-lg text-amber-300 font-mono font-bold">

 剩餘 : 75:00

</div>

</div>

<!-- 交卷後的結算看板 -->

<div id="scoreReportCard" class="hidden bg-white border border-slate-200 p-6 rounded-2xl shadow-sm text-center space-y-3">

<h2 id="resultTitle" class="text-lg font-bold"></h2>

<div class="grid grid-cols-4 gap-3 max-w-lg mx-auto text-center py-2">

<div class="bg-slate-50 p-2.5 rounded-xl border border-slate-200">

總題數

0

</div>

<div class="bg-emerald-50 p-2.5 rounded-xl border border-emerald-200">

答對

0

</div>

<div class="bg-rose-50 p-2.5 rounded-xl border border-rose-200">

答錯

0

</div>

<div class="bg-indigo-50 p-2.5 rounded-xl border border-indigo-200">

得分

0

</div>

</div>

<p class="text-xs text-slate-500">

下方已為您開啟【詳細檢討模式】 : 綠色線為官方正解、紅色線為您的錯誤選擇

，並附帶詳解與權威論文對照。

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</p>
<div class="flex justify-center gap-2 pt-1">
  <button onclick="retryWrongOnly()" id="btnRetryWrong" class="hidden bg-rose-600
hover:bg-rose-700 text-white text-xs font-semibold px-4 py-2 rounded-lg transition">
     僅重練本次錯題 (盲測)
  </button>
  <button onclick="restartExam()" class="bg-slate-100 hover:bg-slate-200 text-slate-700
text-xs font-semibold px-4 py-2 rounded-lg border border-slate-300 transition">
     重新測驗整份試卷
  </button>
</div>
</div>

<!-- 題目列表渲染區 -->
<div id="quizContainer" class="space-y-4"></div>

<!-- 考卷底部常駐交卷欄 -->
<div id="submitBar" class="fixed bottom-0 left-0 right-0 bg-white/95 backdrop-blur-md
border-t border-slate-200 py-3 px-4 z-20">
  <div class="max-w-5xl mx-auto flex items-center justify-between">
    <div class="text-xs text-slate-500">
      已作答 <span id="answeredCount" class="font-bold text-indigo-600">0</span> / <span
id="totalExamCount" class="font-bold">0</span> 題
    </div>
    <button onclick="submitExam()" id="btnSubmitExam" class="bg-indigo-600
hover:bg-indigo-700 text-white font-bold text-xs px-6 py-2.5 rounded-xl shadow-sm transition
cursor-pointer">
       完成作答，交卷看成績與檢討
    </button>
  </div>
</div>
</div>
</section>

<!-- ===== TAB 2: 錯題盲測重練 ===== -->
<section id="tabContent-retry" class="hidden space-y-4">
  <div class="bg-rose-900 text-white p-4 rounded-2xl shadow-sm flex items-center
justify-between">
    <div>
      <h2 class="text-sm font-bold flex items-center gap-1.5">
        <span></span> 錯題盲測重練考場
      </h2>
      <p class="text-xs text-rose-200 mt-0.5">歷史答案已遮蔽。重新獨立思考作答，答對後可
```

一鍵移出錯題紀錄。</p>

</div>

<div class="text-xs font-bold bg-rose-950 px-3 py-1.5 rounded-lg border border-rose-800">

累積錯題: 0 題

</div>

</div>

<div id="retryQuizContainer" class="space-y-4"></div>

</section>

<!-- ===== TAB 3: 5 維度核心概念庫 ===== -->

<section id="tabContent-concepts" class="hidden space-y-4">

<div class="bg-white p-4 rounded-2xl border border-slate-200 shadow-xs space-y-3">

<div class="flex flex-wrap items-center justify-between gap-3">

<div>

<h2 class="text-sm font-bold text-slate-900 flex items-center gap-1.5">

📖 IPAS 必考核心概念庫

</h2>

<p class="text-xs text-slate-500">涵蓋概念分析、底層架構、專有名詞、實戰手法與權威學術論文出處</p>

</div>

<div class="w-72">

<input type="text" id="conceptSearchInput" oninput="filterConcepts()" placeholder="搜尋概念、名詞、論文關鍵字..." class="w-full text-xs border border-slate-300 rounded-lg px-3 py-1.5 focus:outline-none focus:ring-2 focus:ring-indigo-500" />

</div>

</div>

<div class="flex flex-wrap gap-1.5 text-xs pt-1 border-t border-slate-100" id="conceptCategoryBar">

<button class="c-tag active bg-indigo-600 text-white px-2.5 py-1 rounded-lg font-medium" onclick="filterConceptCat('all', this)">全部領域</button>

<button class="c-tag bg-slate-100 text-slate-600 px-2.5 py-1 rounded-lg font-medium hover:bg-slate-200" onclick="filterConceptCat('機器學習基礎', this)">機器學習基礎</button>

<button class="c-tag bg-slate-100 text-slate-600 px-2.5 py-1 rounded-lg font-medium hover:bg-slate-200" onclick="filterConceptCat('深度學習架構', this)">深度學習架構</button>

<button class="c-tag bg-slate-100 text-slate-600 px-2.5 py-1 rounded-lg font-medium hover:bg-slate-200" onclick="filterConceptCat('提示工程與生成式AI', this)">提示工程與生成式AI</button>

<button class="c-tag bg-slate-100 text-slate-600 px-2.5 py-1 rounded-lg font-medium hover:bg-slate-200" onclick="filterConceptCat('RAG與向量系統', this)">RAG與向量系統</button>

<button class="c-tag bg-slate-100 text-slate-600 px-2.5 py-1 rounded-lg font-medium

```
hover:bg-slate-200" onclick="filterConceptCat('AI倫理法規', this)">AI倫理法規</button>
</div>
</div>
```

```
<div id="conceptListContainer" class="space-y-4"></div>
</section>
```

```
<!-- ===== TAB 4: 3D 記憶翻牌卡 (Flashcards)
===== -->
```

```
<section id="tabContent-flashcards" class="hidden space-y-5">
  <div class="bg-indigo-900 text-white p-4 rounded-2xl shadow-sm flex items-center
justify-between">
    <div>
      <h2 class="text-sm font-bold flex items-center gap-1.5">
        <span><img alt="3D flashcard icon" data-bbox="248 358 271 373"/> 3D 記憶翻轉卡 (主動回憶強化)</span>
      </h2>
      <p class="text-xs text-indigo-200 mt-0.5">點擊卡片正面可翻轉查看底層架構、名詞解析
與論文出處，標記熟練度進行間隔複習。</p>
    </div>
    <div class="text-xs font-medium">
      掌握進度：<span id="fcMasteredCount" class="text-emerald-400 font-bold">0</span> /
<span id="fcTotalCount">0</span> 張
    </div>
  </div>
```

```
<div class="max-w-xl mx-auto">
  <div id="flashcardBox" onclick="flipCard()" class="perspective-1000 w-full h-80
cursor-pointer select-none">
    <div id="flashcardInner" class="relative w-full h-full transform-style-3d
transition-transform duration-500 rounded-2xl shadow-md border border-slate-200 bg-white">
      <!-- 正面 -->
      <div class="absolute inset-0 backface-hidden p-6 flex flex-col justify-between
rounded-2xl bg-white">
        <div class="flex justify-between items-center text-xs text-slate-400">
          <span id="fcCategory" class="bg-indigo-50 text-indigo-700 font-semibold px-2 py-0.5
rounded">分類</span>
          <span id="fcIndex" class="font-mono">1 / 10</span>
        </div>
        <div class="my-auto text-center px-4">
          <span class="text-xs text-slate-400 block mb-2 font-medium">【核心考點問題】
</span>
          <h3 id="fcFrontText" class="text-base font-bold text-slate-900 leading-relaxed">問題
載入中...</h3>
```

```

</div>
<div class="text-center text-slate-400 text-[11px]">
  🖱️ 點擊卡片翻轉看答案與架構
</div>
</div>

<!-- 背面 -->
<div class="absolute inset-0 backface-hidden rotate-y-180 p-6 flex flex-col
justify-between rounded-2xl bg-slate-900 text-white overflow-y-auto">
  <div class="flex justify-between items-center text-xs text-slate-400 border-b
border-slate-800 pb-2">
    <span class="text-emerald-400 font-semibold">💡 核心解析與記憶重點</span>
    <span class="text-[10px] text-slate-400">點擊任意處翻回</span>
  </div>
  <div class="my-auto space-y-2 py-2 text-xs">
    <p id="fcBackAnswer" class="text-slate-200 leading-relaxed font-medium"></p>
    <div class="p-2.5 rounded bg-slate-800/80 border border-slate-700 text-[11px]
text-indigo-300">
      <span class="font-bold block text-indigo-400">底層機制 / 手法 : </span>
      <span id="fcBackMethod"></span>
    </div>
    <div class="text-[10px] text-slate-400">
      📄 權威論文出處 : <span id="fcBackPaper" class="text-slate-300
underline"></span>
    </div>
  </div>
  <div class="text-center text-slate-500 text-[10px]">
    點擊翻回正面
  </div>
</div>
</div>

<div class="flex items-center justify-between mt-4">
  <button onclick="prevFlashcard()" class="px-3 py-1.5 text-xs bg-white border
border-slate-200 rounded-lg hover:bg-slate-50 text-slate-700">
    ⬅️ 上一張
  </button>
  <div class="flex gap-2">
    <button onclick="markCardMastery(false)" class="px-3 py-1.5 text-xs bg-rose-50
border border-rose-200 text-rose-700 font-medium rounded-lg hover:bg-rose-100">
      😊 還不熟
    </button>
  </div>
</div>

```

```
<button onclick="markCardMastery(true)" class="px-3 py-1.5 text-xs bg-emerald-50 border border-emerald-200 text-emerald-700 font-medium rounded-lg hover:bg-emerald-100">
```

✅ 已記住

```
</button>
```

```
</div>
```

```
<button onclick="nextFlashcard()" class="px-3 py-1.5 text-xs bg-white border border-slate-200 rounded-lg hover:bg-slate-50 text-slate-700">
```

下一張 ➡

```
</button>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- ===== TAB 5: 做題分析與弱點診斷 ===== -->
```

```
<section id="tabContent-analytics" class="hidden space-y-5">
```

```
<div class="grid grid-cols-4 gap-3 text-center">
```

```
<div class="bg-white p-4 rounded-2xl border border-slate-200 shadow-xs">
```

```
<span class="text-slate-400 text-[11px] block">作答題數</span>
```

```
<span id="anTotal" class="text-xl font-bold text-slate-900">0</span>
```

```
</div>
```

```
<div class="bg-emerald-50 p-4 rounded-2xl border border-emerald-200 shadow-xs">
```

```
<span class="text-emerald-600 text-[11px] block">答對題數</span>
```

```
<span id="anCorrect" class="text-xl font-bold text-emerald-700">0</span>
```

```
</div>
```

```
<div class="bg-rose-50 p-4 rounded-2xl border border-rose-200 shadow-xs">
```

```
<span class="text-rose-600 text-[11px] block">答錯題數</span>
```

```
<span id="anWrong" class="text-xl font-bold text-rose-700">0</span>
```

```
</div>
```

```
<div class="bg-indigo-50 p-4 rounded-2xl border border-indigo-200 shadow-xs">
```

```
<span class="text-indigo-600 text-[11px] block">整體正確率</span>
```

```
<span id="anAccuracy" class="text-xl font-bold text-indigo-700">0%</span>
```

```
</div>
```

```
</div>
```

```
<div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-xs space-y-3">
```

```
<div class="flex justify-between items-center border-b border-slate-100 pb-2">
```

```
<h3 class="font-bold text-xs text-slate-900 flex items-center gap-1.5">
```

```
<span>📊</span> 各考點領域掌握度分佈
```

```
</h3>
```

```
<span class="text-[10px] text-slate-400">綠色 ≥75% 精通 / 黃色 60-74% / 紅色 <60%
```

```
弱項</span>
```

```
</div>
```

```
<div id="categoryBars" class="space-y-2.5 text-xs">
```

```
        <div class="text-slate-400 text-center py-6">尚未有作答紀錄，請於考場作答交卷後查看！</div>
    </div>
</div>
```

```
    <div class="border border-indigo-200 bg-indigo-50/50 p-5 rounded-2xl space-y-3">
        <div class="flex items-center justify-between">
            <h3 class="font-bold text-xs text-indigo-950 flex items-center gap-1.5">
                <span><img alt="AI icon" data-bbox="248 243 271 260"/> AI 考前弱點處方籤與複習指引</span>
            </h3>
            <button onclick="generateAIWeaknessReport()" id="btnGenAIReport"
class="bg-indigo-600 hover:bg-indigo-700 text-white px-3 py-1.5 rounded-lg text-xs
font-semibold transition">
                一鍵生成弱點診斷
            </button>
        </div>
        <div id="aiWeaknessContent" class="text-slate-700 leading-relaxed whitespace-pre-wrap
text-xs bg-white p-4 rounded-xl border border-indigo-100 min-h-[80px]">
            點擊右上角按鈕，AI 助教會根據你交卷後的錯題章節與答對率，為你量身打造考前 7 天
專屬補強清單。
        </div>
    </div>
```

```
    <div class="text-right pt-2">
        <button onclick="resetAllStats()" class="text-rose-600 hover:text-rose-800 text-xs
font-medium underline">
            <img alt="trash icon" data-bbox="178 583 201 600"/> 清除所有做題與錯題紀錄重新開始
        </button>
    </div>
</section>
```

```
</div>
</div>
```

```
<!-- ===== 4. 右下角常駐: AI 即時聯網助教 ===== -->
<div class="fixed bottom-16 right-6 z-40">
    <button onclick="toggleChatDrawer()" class="bg-indigo-600 hover:bg-indigo-700 text-white
p-3.5 rounded-full shadow-lg flex items-center gap-2 transition hover:scale-105">
        ✨ <span class="text-xs font-semibold pr-1">AI 聯網助教</span>
    </button>
</div>
```

```
<!-- AI 即時對話抽屜 -->
```

```
<div id="chatDrawer" class="fixed bottom-28 right-6 w-96 max-w-[calc(100vw-3rem)] h-[500px]
bg-white rounded-2xl shadow-2xl border border-slate-200 z-50 flex flex-col hidden
overflow-hidden text-xs">
```

```
<div class="bg-indigo-600 text-white px-4 py-3 flex items-center justify-between">
```

```
<div>
```

```
<h4 class="text-xs font-bold leading-tight flex items-center gap-1"><img alt="AI icon" data-bbox="688 188 708 203"/> AI 聯網即時助教
</h4>
```

```
<span class="text-[10px] text-indigo-200"><img alt="Google logo" data-bbox="198 225 218 240"/> 具備 Google 即時搜尋與論文溯源</span>
```

```
</div>
```

```
<button onclick="toggleChatDrawer()" class="text-indigo-200 hover:text-white font-bold
text-sm">X </button>
```

```
</div>
```

```
<div id="chatMessages" class="flex-1 p-4 overflow-y-auto space-y-3 bg-slate-50">
```

```
<div class="bg-white border border-slate-200 p-3 rounded-xl shadow-xs text-slate-700">
```

```
你好！我是你的 iPAS AI 備考即時助教。你可以隨時問我：
```

```
<ul class="list-disc pl-4 mt-1 space-y-0.5 text-slate-500">
```

```
<li>機器學習與深度學習核心公式與概念</li>
```

```
<li>檢索開創性論文出處（如 Transformer、RAG、LoRA）</li>
```

```
<li>詢問歷屆容易混淆的單選陷阱</li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
<div class="p-3 bg-white border-t border-slate-200">
```

```
<div class="flex items-center gap-2">
```

```
<input type="text" id="chatInput" placeholder="輸入問題（如：查詢 RAG 的開創論文）..."
onkeydown="if(event.key==='Enter') sendChat()" class="flex-1 text-xs border border-slate-300
rounded-lg px-3 py-2 focus:outline-none focus:ring-2 focus:ring-indigo-500" />
```

```
<button onclick="sendChat()" class="bg-indigo-600 hover:bg-indigo-700 text-white px-3
py-2 rounded-lg text-xs font-bold">送出</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- API Key 設定彈窗 -->
```

```
<div id="keyModal" class="fixed inset-0 bg-black/40 backdrop-blur-xs z-50 flex items-center
justify-center hidden p-4">
```

```
<div class="bg-white rounded-xl max-w-md w-full p-6 shadow-xl border border-slate-200">
```

```
<h3 class="text-base font-bold text-slate-900 mb-2"><img alt="Key icon" data-bbox="565 828 585 843"/> 設定 Gemini API Key</h3>
```

```
<p class="text-xs text-slate-500 mb-4 leading-relaxed">
```

```
可至 <a href="https://aistudio.google.com/" target="_blank" class="text-indigo-600
underline font-semibold">Google AI Studio</a> 免費取得 API Key。金鑰僅儲存在本地瀏覽器。
```

```

    </p>
    <input type="password" id="geminiApiKeyInput" placeholder="貼上 API Key..." class="w-full
text-xs border border-slate-300 rounded-lg px-3 py-2 focus:outline-none focus:ring-2
focus:ring-indigo-500 mb-4" />
    <div class="flex justify-end space-x-2">
        <button onclick="closeKeySettings()" class="px-3 py-1.5 text-xs text-slate-600
hover:bg-slate-100 rounded-lg">取消</button>
        <button onclick="saveKeySettings()" class="px-4 py-1.5 text-xs bg-indigo-600 text-white
rounded-lg hover:bg-indigo-700">儲存</button>
    </div>
</div>
</div>

```

<!-- ===== 5. 核心 JavaScript 邏輯 ===== -->

```

<script>
// 預設通行密碼(後台驗證常數, 前台介面無任何明文顯示)
const CORRECT_PASSWORD = "ipas2026";
let currentCaptcha = "";
let isMemoryAuthed = false;

// 安全 Storage 操作 (全面相容 Safari 無痕模式)
function safeStorageGet(key) {
    try { return sessionStorage.getItem(key); } catch (e) { return null; }
}
function safeStorageSet(key, val) {
    try { sessionStorage.setItem(key, val); } catch (e) {}
}
function safeStorageRemove(key) {
    try { sessionStorage.removeItem(key); } catch (e) {}
}

function generateCaptcha() {
    const chars = "23456789ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    let code = "";
    for (let i = 0; i < 4; i++) {
        code += chars.charAt(Math.floor(Math.random() * chars.length));
    }
    return code;
}

function refreshCaptcha() {
    currentCaptcha = generateCaptcha();
    const el = document.getElementById("captchaDisplay");

```

```
    if (el) el.innerText = currentCaptcha;
}
```

```
function checkAuthStatus() {
    const isAuthenticated = safeStorageGet("IPAS_APP_AUTH") === "true" || isMemoryAuthenticated;
    const gate = document.getElementById("authGate");
    const app = document.getElementById("mainApp");

    if (isAuthenticated) {
        gate.classList.add("hidden");
        app.classList.remove("hidden");
        initApp();
    } else {
        gate.classList.remove("hidden");
        app.classList.add("hidden");
        refreshCaptcha();
    }
}
```

```
function handleUnlock() {
    const pwdInput = document.getElementById("inputPassword");
    const captchaInput = document.getElementById("inputCaptcha");
    const err = document.getElementById("authErrorMsg");

    const pwd = pwdInput ? pwdInput.value.trim() : "";
    const captcha = captchaInput ? captchaInput.value.trim().toUpperCase() : "";
```

```
    if (pwd !== CORRECT_PASSWORD) {
        err.innerText = "通行密碼錯誤，請重新輸入！";
        err.classList.remove("hidden");
        refreshCaptcha();
        if (captchaInput) captchaInput.value = "";
        return;
    }
```

```
    if (captcha !== currentCaptcha) {
        err.innerText = "驗證碼錯誤，請輸入右側顯示的 4 碼！";
        err.classList.remove("hidden");
        refreshCaptcha();
        if (captchaInput) captchaInput.value = "";
        return;
    }
```

```

err.classList.add("hidden");
safeStorageSet("IPAS_APP_AUTH", "true");
isMemoryAuthed = true;
checkAuthStatus();
}

function lockApp() {
  safeStorageRemove("IPAS_APP_AUTH");
  isMemoryAuthed = false;
  checkAuthStatus();
}

// ===== 模組分頁切換 =====
function switchTab(tabId) {
  const tabs = ["quiz", "retry", "concepts", "flashcards", "analytics"];
  tabs.forEach(t => {
    const content = document.getElementById(`tabContent-${t}`);
    const btn = document.getElementById(`tabBtn-${t}`);
    if (t === tabId) {
      content.classList.remove("hidden");
      btn.classList.add("bg-white", "text-indigo-600", "shadow-xs");
      btn.classList.remove("text-slate-600");
    } else {
      content.classList.add("hidden");
      btn.classList.remove("bg-white", "text-indigo-600", "shadow-xs");
      btn.classList.add("text-slate-600");
    }
  });

  if (tabId === "retry") renderRetryQuestions();
  if (tabId === "concepts") renderConcepts();
  if (tabId === "flashcards") renderFlashcard();
  if (tabId === "analytics") updateAnalyticsView();
}

// ===== 核心題庫(預設 10 道標準全真考題) =====
const initialExamQuestions = [
  {
    id: "1",
    subject: "初級科目一",
    category: "機器學習基礎",
    question: "在機器學習任務中, 若希望根據房屋的坪數、屋齡與地段來預測其成交價格(連續數值), 應採用下列何種模型?",
  }

```

options: { "A": "分群 (Clustering)", "B": "迴歸 (Regression)", "C": "關聯規則 (Association Rule)", "D": "主成分分析 (PCA)" },

answer: ["B"],

explanation: "預測連續數值目標屬於監督式學習中的『迴歸』任務;分類任務預測離散類別,分群與降維則多為非監督式學習。",

paper: "Hastie et al. - 《The Elements of Statistical Learning》"

},

{

id: "2",

subject: "初級科目一",

category: "模型評估指標",

question: "在醫療診斷或罕見疾病篩檢情境中, 最希望避免『有病卻被誤判為沒病(漏診)』的情況, 此時最應優先追求哪一項評估指標的極大化?",

options: { "A": "精確率 (Precision)", "B": "召回率 (Recall / Sensitivity)", "C": "特異度 (Specificity)", "D": "準確率 (Accuracy)" },

answer: ["B"],

explanation: "漏診代表偽陰性 (False Negative, FN)。Recall = TP / (TP + FN), 當 FN 越小, Recall 越高。因此避免漏診時應優先提升 Recall。",

paper: "Davis & Goadrich, ICML 2006 - 《The Relationship Between Precision-Recall and ROC Curves》"

},

{

id: "3",

subject: "初級科目一",

category: "過擬合與正規化",

question: "當神經網路在訓練集上表現極佳, 但在測試集上泛化能力很差時, 下列哪項手法『無法』有效減緩過擬合?",

options: { "A": "加入 Dropout 機制", "B": "加入 L2 正則化 (Weight Decay)", "C": "大幅增加網路隱藏層的神經元數量與深度", "D": "增加訓練資料多樣性 (Data Augmentation)" },

answer: ["C"],

explanation: "大幅增加神經元數量與深度會提高模型容量 (Capacity), 反而會加劇過擬合。其餘三項皆為抑制過擬合的標準手段。",

paper: "Srivastava et al., JMLR 2014 - 《Dropout: A Simple Way to Prevent Overfitting》"

},

{

id: "4",

subject: "初級科目二",

category: "RAG與向量系統",

question: "企業在導入大型語言模型 (LLM) 建置內部知識庫時, 為了降低『幻覺 (Hallucination)』並補充內部最新私有資料, 最常採取的低成本架構為何?",

options: { "A": "從頭重新預訓練基底大模型", "B": "檢索增強生成 (Retrieval-Augmented Generation, RAG)", "C": "大幅調高模型的 Temperature 參數", "D": "使用全參數量微調 (Full Fine-Tuning)" },

```
    answer: ["B"],
    explanation: "RAG 透過向量比對先從內部庫取得文件片段作為 Context 注入 Prompt, 由 LLM 整理回答, 成本低且能大幅緩解幻覺。",
    paper: "Lewis et al., NeurIPS 2020 - 《Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks》"
  },
  {
    id: "5",
    subject: "初級科目二",
    category: "提示工程與生成式AI",
    question: "在 Prompt Engineering 中, 要求模型在給出答案前先輸出中間推導步驟 (如:『請一步一步思考』), 此手法稱為?",
    options: { "A": "Zero-shot Prompting", "B": "思維鏈 (Chain-of-Thought, CoT)", "C": "Prompt 注入攻擊", "D": "量化壓縮 (Quantization)" },
    answer: ["B"],
    explanation: "思維鏈 (CoT) 引導模型展開中間步驟推理, 能顯著提升複雜邏輯與算術題目的正確率。",
    paper: "Wei et al., NeurIPS 2022 - 《Chain-of-Thought Prompting Elicits Reasoning in Large Language Models》"
  },
  {
    id: "6",
    subject: "初級科目二",
    category: "模型參數調校",
    question: "在呼叫生成式 AI API 時, 若希望模型生成的回答具備更高的創造力與多樣性, 通常應如何調整 Temperature 參數?",
    options: { "A": "調低接近 0 (如 0.1)", "B": "調高 (如 0.8 ~ 1.0)", "C": "將 Temperature 設為負數", "D": "Temperature 與創造力無關" },
    answer: ["B"],
    explanation: "Temperature 決定輸出機率分佈平滑度。越接近 0 越偏向 Greedy 確定性輸出; 調高會使低機率 Token 更有機會被選中, 增加創造力與隨機性。",
    paper: "Ackley et al., 1985 - 《A Learning Algorithm for Boltzmann Machines》"
  },
  {
    id: "7",
    subject: "初級科目一",
    category: "特徵工程與資料前處理",
    question: "在特徵縮放 (Feature Scaling) 中, 將數值特徵線性變換縮放至 [0, 1] 區間的方法稱為?",
    options: { "A": "Z-score 標準化 (Standardization)", "B": "最小最大值正規化 (Min-Max Normalization)", "C": "主成分分析 (PCA)", "D": "One-Hot 編碼" },
    answer: ["B"],
    explanation: "Min-Max Normalization 公式為  $(X - X_{min}) / (X_{max} - X_{min})$ , 將資料線性對齊
```

至 [0, 1]; Z-score 是將平均數轉為 0、標準差轉為 1。",
 paper: "Scikit-Learn Preprocessing Documentation"
 },
 {
 id: "8",
 subject: "初級科目二",
 category: "AI倫理法規",
 question: "依據台灣國科會《人工智慧基本法(草案)》與數位發展部公部門 AI 應用指引, 下列哪項敘述為正確的核心原則?",
 options: { "A": "公務機關可直接將高度機密公文上傳至未授權之公有雲 AI 模型分析", "B": "AI 系統產出之內容應具備透明性與可解釋性, 並由人類負最終決策責任", "C": "生成式 AI 生成之內容具有絕對法律權利, 可直接列為專利發明人", "D": "導入 AI 應完全取代人類審查, 不得保留人工覆核環節" },
 answer: ["B"],
 explanation: "AI 倫理與各國指引一致強調 Human-in-the-loop(人機協作/人為最終問責), AI 不具備法律主體專利人地位, 機密公文亦嚴禁上傳未授權公有雲。",
 paper: "台灣國科會《人工智慧基本法(草案)》& NIST AI RMF 1.0"
 },
 {
 id: "9",
 subject: "初級科目一",
 category: "深度學習架構",
 question: "在卷積神經網路(CNN)中, 『池化層(Pooling Layer, 如 Max Pooling)』的主要功能為何?",
 options: { "A": "計算損失函數的梯度", "B": "降低特徵維度、減少參數量並保留主要特徵", "C": "將連續數值轉換為文字標籤", "D": "取代 Softmax 進行最後分類" },
 answer: ["B"],
 explanation: "池化層用於降低特徵圖(Feature Map)空間維度, 減少計算量與防止過擬合, 同時具備一定的平移不變性。",
 paper: "LeCun et al., 1998 - 《Gradient-Based Learning Applied to Document Recognition》"
 },
 {
 id: "10",
 subject: "初級科目二",
 category: "模型微調技術",
 question: "在大型語言模型的高效微調技術中, LoRA(Low-Rank Adaptation)的核心原理為何?",
 options: { "A": "凍結預訓練權重, 透過引入低秩矩陣分解來訓練額外參數", "B": "將模型參數量修剪掉 90% 來提升推論速度", "C": "直接在提示詞前端拼接人工撰寫的樣板", "D": "將模型權重全數轉為 1-bit 二值化" },
 answer: ["A"],
 explanation: "LoRA 凍結原有大模型權重 W, 透過旁路學習兩個低秩分解矩陣 A 與 B($\Delta W = B \times A$), 大幅降低微調所需的 GPU 顯存與可訓練參數量。",

```

    paper: "Hu et al., ICLR 2022 - 《LoRA: Low-Rank Adaptation of Large Language Models》"
  }
];

// ===== 5 維度核心概念庫資料庫 =====
const conceptDatabase = [
  {
    id: "rag_architecture",
    title: "檢索增強生成 (RAG)",
    category: "RAG與向量系統",
    analysis: "解決大型語言模型 (LLM) 無法即時更新內部權重、無法取得企業內部私有資料以及產生事實幻覺 (Hallucination) 三大難題。在不修改模型權重的前提下外掛專屬知識庫。",
    pipeline: "【文本資料切塊 Chunking】→【向量模型轉換 Embeddings】→【儲存至向量庫 Vector DB】→【提問比對 Top-K 檢索】→【上下文注入 Context 擴增提示】→【LLM 生成精準回答】",
    glossary: "Chunk Overlap (重疊切分防止斷意)、Cosine Similarity (餘弦相似度比對)、Embedding Dimension (維度空間)、Top-K (選取最相關文件篇數)、Temperature (解碼發散度控制)。",
    techniques: "• Hybrid Search: 結合傳統 BM25 關鍵字搜尋與向量語意搜尋。\\n• Re-ranking: 檢索後透過專用 Cross-Encoder 重排序提升精準度。\\n• Parent-Child Chunking: 檢索小區塊, 回填所屬完整大章節上下文。",
    paper: "Lewis et al., NeurIPS 2020 - 《Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks》",
    paper_url: "https://arxiv.org/abs/2005.11401"
  },
  {
    id: "prompt_cot",
    title: "思維鏈提示 (Chain-of-Thought, CoT)",
    category: "提示工程與生成式AI",
    analysis: "傳統 Prompt 直接要求模型給予最終答案, 遇到多步驟邏輯或數學題極易出錯。CoT 引導模型把複雜問題拆解為步驟依序推理。",
    pipeline: "【輸入問題】→【激發提示 (如: Let's think step by step)】→【逐步產生中間推論 Token】→【依據推論路徑輸出最終結論】",
    glossary: "Zero-shot CoT (直接加入口訣觸發推理)、Few-shot CoT (給予包含推導過程的範例)、ToT (Tree of Thoughts 思維樹分支推演)。",
    techniques: "• 角色扮演法 (System Role): 先賦予專業專家身分。\\n• 結構化格式規範: 強制要求以 JSON 或 Markdown 列點逐步推演。\\n• 自我一致性採樣 (Self-Consistency): 多路徑推演取眾數答案。",
    paper: "Wei et al., NeurIPS 2022 - 《Chain-of-Thought Prompting Elicits Reasoning in Large Language Models》",
    paper_url: "https://arxiv.org/abs/2201.11903"
  },
  {

```

```

id: "transformer_attention",
title: "Transformer 與自注意力機制",
category: "深度學習架構",
analysis: "徹底取代傳統 RNN 無法平行運算與容易遺忘長期依賴的缺點，是當代所有大語言模型 (GPT、Claude、Gemini) 與大視覺模型的共同底層架構。",
pipeline: "【輸入文字 Embedding】→【注入位置編碼 Positional Encoding】→【Q(Query), K(Key), V(Value) 矩陣點積注意力】→【Multi-Head 多頭投影】→【Feed-Forward 前饋網路與殘差連接】",
glossary: "Self-Attention (自注意力權重計算)、Multi-Head Attention (多角度特徵擷取)、Softmax 權重歸一化、Residual Connection (殘差連接解決梯度消失)。",
techniques: "• Scaling Factor (除以根號 dk 防止 Softmax 梯度飽和)\n• Causal Masking (在解碼器中遮蔽未來 Token 防止答案洩漏)\n• FlashAttention (GPU 記憶體存取優化)",
paper: "Vaswani et al., NeurIPS 2017 - 《Attention Is All You Need》",
paper_url: "https://arxiv.org/abs/1706.03762"
},
{
id: "model_overfitting",
title: "過擬合治理與正規化 (Regularization)",
category: "機器學習基礎",
analysis: "當模型在訓練集表現近乎完美，但在未見過的測試集上表現低落，即為高變異數 (High Variance) 的過擬合現象。模型過度記憶雜訊。",
pipeline: "【監測驗證集損失 Validation Loss】→【若訓練損失下降但驗證損失上升】→【啟動正則化懲罰 / Dropout / 提早停止機制】",
glossary: "Bias-Variance Tradeoff (偏差變異數權衡)、L1 正則化 (Lasso, 產生稀疏權重)、L2 正則化 (Ridge, 權重衰減 Weight Decay)、Dropout (隨機拋棄神經節點)。",
techniques: "• Data Augmentation: 翻轉、旋轉、加雜訊擴增訓練資料。\n• Early Stopping: 當驗證損失連續 N 個 Epoch 未下降即中斷訓練。\n• Dropout 比例設定 (通常為 0.2 ~ 0.5)。",
paper: "Srivastava et al., JMLR 2014 - 《Dropout: A Simple Way to Prevent Neural Networks from Overfitting》",
paper_url: "https://jmlr.org/papers/v15/srivastava14a.html"
},
{
id: "ai_governance",
title: "AI 倫理、治理與法規標準",
category: "AI 倫理法規",
analysis: "iPAS 必考高頻題目，強調技術落地時必須兼顧可信任度、公平性、個人資料保護與演算法可解釋性。",
pipeline: "【風險評估與分類】→【數據隱私與去識別化】→【演算法偏見審查 Bias Audit】→【人機協作 Human-in-the-loop 覆核】",
glossary: "可信任 AI (Trustworthy AI)、數據毒化 (Data Poisoning)、演算法偏見 (Algorithmic Bias)、版權與合理使用 (Fair Use)。",
techniques: "• 採用 ISO/IEC 42001 人工智慧管理系統作為內部合規標準。\n• 遵循台灣數位發展部《公部門 AI 應用參考手冊》規範公文書產出。\n• 敏感個人資料嚴禁直接傳送至公有雲未

```

授權模型。",

paper: "NIST AI RMF 1.0 (2023) & 台灣國科會《人工智慧基本法(草案)》",

paper_url: "https://www.nist.gov/itl/ai-risk-management-framework"

}

];

let examQuestions = [...initialExamQuestions];

let userAnswers = {}; // { qId: ["A"] }

let isSubmitted = false;

let lastWrongQuestions = [];

let wrongQuestions = [];

let topicAnalytics = {};

let flashcardList = [];

let currentFcIndex = 0;

let timerInterval = null;

try {

wrongQuestions = JSON.parse(localStorage.getItem("IPAS_WRONG_QUESTIONS") || "[]");

topicAnalytics = JSON.parse(localStorage.getItem("IPAS_TOPIC_STATS") || "{}");

} catch(e) {}

function initApp() {

updateWrongBadge();

updateSummary();

updateKeyStatus();

initFlashcards();

restartExam();

}

function updateSummary() {

document.getElementById("quizCountSummary").innerText = `全真題庫

\${examQuestions.length} 題 · 核心概念庫 \${conceptDatabase.length} 篇`;

}

function updateWrongBadge() {

document.getElementById("navWrongCount").innerText = wrongQuestions.length;

const rTot = document.getElementById("retryWrongTotal");

if (rTot) rTot.innerText = wrongQuestions.length;

}

// 重新開始試卷

function restartExam() {

isSubmitted = false;

```

userAnswers = {};
document.getElementById("scoreReportCard").classList.add("hidden");
document.getElementById("submitBar").classList.remove("hidden");
document.getElementById("totalExamCount").innerText = examQuestions.length;
updateAnsweredCount();
renderExamQuestions();
window.scrollTo({ top: 0, behavior: "smooth" });
}

// 渲染試卷全部題目 (作答時絕不劇透)
function renderExamQuestions() {
  const container = document.getElementById("quizContainer");
  container.innerHTML = "";

  examQuestions.forEach((q, idx) => {
    const card = document.createElement("div");
    card.id = `qCard-${q.id}`;
    card.className = "bg-white border border-slate-200 rounded-2xl p-5 shadow-xs transition";

    const isMulti = q.type === "multiple" || q.answer.length > 1;
    const inputType = isMulti ? "checkbox" : "radio";
    const userSelected = userAnswers[q.id] || [];

    let optionsHtml = Object.entries(q.options).map(([optKey, optVal]) => {
      const isChecked = userSelected.includes(optKey);
      let optStyle = "border-slate-200 hover:bg-slate-50";

      // 交卷後紅綠對照線
      if (isSubmitted) {
        const isCorrectOpt = q.answer.includes(optKey);
        if (isCorrectOpt) {
          optStyle = "bg-emerald-50 border-emerald-500 text-emerald-950 font-bold ring-1 ring-emerald-400";
        } else if (isChecked && !isCorrectOpt) {
          optStyle = "bg-rose-50 border-rose-500 text-rose-950 line-through ring-1 ring-rose-400";
        }
      }
    });

    return `
      <label class="flex items-start p-3 border rounded-xl cursor-pointer transition
        ${optStyle}">

```

```



```

```

card.innerHTML = `
<div class="flex items-center justify-between text-xs text-slate-400 mb-2">
  <span class="bg-indigo-50 text-indigo-700 font-bold px-2 py-0.5 rounded">第 ${idx + 1}
題 · ${q.subject || '考題'}</span>
  <span>${isMulti ? '【多選題】' : '【單選題】'} · ${q.category || ""}</span>
</div>

```

```

<p class="text-sm font-medium text-slate-900 mb-4 leading-normal">${q.question}</p>
<div class="space-y-2 mb-3">${optionsHtml}</div>

```

```

<!-- 交卷後顯示詳解、觀念卡與論文出處 -->
<div id="feedback-${q.id}" class="${isSubmitted ? '' : 'hidden'} mt-4 pt-3 border-t
border-slate-100 space-y-2 text-xs">
  <div class="p-3 bg-slate-50 border border-slate-200 rounded-xl text-slate-700
leading-relaxed">
    <span class="font-bold block text-slate-900 mb-1">💡 官方詳解:</span>
    ${q.explanation || '暫無解析'}
  </div>
  ${q.paper ? `
    <div class="p-2.5 bg-indigo-50/60 border border-indigo-100 rounded-xl text-indigo-900
leading-relaxed text-[11px]">
      <span class="font-bold">📖 權威論文 / 規範出處:</span> ${q.paper}
    </div>` : ''}
  <div class="flex justify-between items-center pt-1">
    <button onclick="addQuestionToFlashcard('${q.id}')" class="text-xs text-slate-500
hover:text-indigo-600">
      🔄 轉為記憶翻牌卡

```

```

        </button>
        <button onclick="askQuestionAI('${q.id}')" class="text-indigo-600 hover:underline
text-xs font-semibold">
            ✨ 請 AI 助教深入解說此題
        </button>
    </div>
    <div id="aiReply-${q.id}" class="hidden p-3 bg-indigo-50 border border-indigo-200
rounded-xl text-indigo-950 whitespace-pre-wrap leading-relaxed text-xs"></div>
</div>
`;

```

```

        container.appendChild(card);
    });
}

```

```

function handleSelectOption(qId, optKey, isMulti) {
    if (isSubmitted) return;

```

```

    if (isMulti) {
        if (!userAnswers[qId]) userAnswers[qId] = [];
        const idx = userAnswers[qId].indexOf(optKey);
        if (idx > -1) userAnswers[qId].splice(idx, 1);
        else userAnswers[qId].push(optKey);
    } else {
        userAnswers[qId] = [optKey];
    }

```

```

    updateAnsweredCount();
}

```

```

function updateAnsweredCount() {
    const answered = Object.values(userAnswers).filter(ans => ans.length > 0).length;
    document.getElementById("answeredCount").innerText = answered;
}

```

```

// 交卷計算成績與展開紅線對照線

```

```

function submitExam() {
    const answered = Object.values(userAnswers).filter(ans => ans.length > 0).length;
    const total = examQuestions.length;

```

```

    if (answered < total) {
        if (!confirm(`您還有 ${total - answered} 題尚未作答，確定現在交卷嗎？`)) {
            return;

```

```
}  
}
```

```
isSubmitted = true;  
document.getElementById("submitBar").classList.add("hidden");
```

```
let correctCount = 0;  
lastWrongQuestions = [];
```

```
examQuestions.forEach(q => {  
  const userSelected = (userAnswers[q.id] || []).sort();  
  const correctAns = [...q.answer].sort();  
  const isCorrect = JSON.stringify(userSelected) === JSON.stringify(correctAns);
```

```
  // 累積分析紀錄  
  const cat = q.category || "綜合領域";  
  if (!topicAnalytics[cat]) topicAnalytics[cat] = { total: 0, correct: 0 };  
  topicAnalytics[cat].total++;
```

```
  if (isCorrect) {  
    correctCount++;  
    topicAnalytics[cat].correct++;  
  } else {  
    lastWrongQuestions.push(q);  
    if (!wrongQuestions.some(item => item.id === q.id)) {  
      wrongQuestions.push(q);  
    }  
  }  
});
```

```
try {  
  localStorage.setItem("IPAS_TOPIC_STATS", JSON.stringify(topicAnalytics));  
  localStorage.setItem("IPAS_WRONG_QUESTIONS", JSON.stringify(wrongQuestions));  
} catch(e) {}  
updateWrongBadge();
```

```
const wrongCount = total - correctCount;  
const score = Math.round((correctCount / total) * 100);  
const isPassed = score >= 70;
```

```
// 顯示成績單  
const card = document.getElementById("scoreReportCard");  
card.classList.remove("hidden");
```

```

document.getElementById("resTotal").innerText = total;
document.getElementById("resCorrect").innerText = correctCount;
document.getElementById("resWrong").innerText = wrongCount;
document.getElementById("resScore").innerText = `${score}分`;

const title = document.getElementById("resultTitle");
if (isPassed) {
  title.className = "text-lg font-bold text-emerald-600";
  title.innerText = `🎉 恭喜及格！成績: ${score} 分 (及格標準 70 分)`;
} else {
  title.className = "text-lg font-bold text-rose-600";
  title.innerText = `測驗未達及格標準。成績: ${score} 分 (及格標準 70 分)`;
}

const btnRetry = document.getElementById("btnRetryWrong");
if (wrongCount > 0) btnRetry.classList.remove("hidden");
else btnRetry.classList.add("hidden");

renderExamQuestions();
window.scrollTo({ top: 0, behavior: "smooth" });
}

// 僅重練本次錯題
function retryWrongOnly() {
  if (lastWrongQuestions.length === 0) return;
  examQuestions = [...lastWrongQuestions];
  restartExam();
}

// 載入官方 790 題庫隨機抽題
async function loadOfficial790() {
  if (!confirm("確定要載入官方 790 題題庫並抽取 20 題進行測驗嗎？")) return;

  try {
    const res = await fetch("https://yazelin.github.io/ipas-ai-quiz/questions.json");
    const data = await res.json();
    const rawList = data.questions || data;

    const letters = ["A", "B", "C", "D", "E"];
    const all790 = rawList.map((raw, idx) => {
      let opts = {};
      if (Array.isArray(raw.options)) {
        raw.options.forEach((opt, i) => { opts[letters[i]] = opt; });
      }
    });
  }
}

```

```

    } else {
      opts = raw.options || {};
    }
    let ans = [];
    if (typeof raw.answer === "number") ans = [letters[raw.answer]];
    else if (Array.isArray(raw.answer)) ans = raw.answer.map(a => typeof a === "number" ?
letters[a] : a);
    else if (typeof raw.answer === "string") ans = [raw.answer.trim().toUpperCase()];

    return {
      id: raw.id || `Q-${idx + 1}`,
      subject: raw.level || raw.subject || "初級",
      category: raw.category || raw.topic || "歷屆試題",
      type: ans.length > 1 ? "multiple" : "single",
      question: raw.question,
      options: opts,
      answer: ans,
      explanation: raw.explanation || "暫無官方解析。",
      paper: "經濟部 iPAS 官方能力鑑定公告試題"
    };
  });

  examQuestions = all790.sort(() => 0.5 - Math.random()).slice(0, 20);
  alert(`成功載入！已為您抽出 20 題模擬試題，開始作答！`);
  restartExam();
} catch(err) {
  alert("載入官方題庫失敗，請確認網路狀態。");
}
}

```

```

function startMockExam(count) {
  if (timerInterval) clearInterval(timerInterval);
  const container = document.getElementById("examTimerContainer");
  const display = document.getElementById("examTimer");
  container.classList.remove("hidden");
  container.classList.add("flex");

  let remain = 75 * 60;
  timerInterval = setInterval(() => {
    remain--;
    const m = String(Math.floor(remain / 60)).padStart(2, '0');
    const s = String(remain % 60).padStart(2, '0');
    display.innerText = `${m}:${s}`;
  }, 1000);
}

```

```

    if (remain <= 0) {
      clearInterval(timerInterval);
      alert("時間到！自動為您交卷。");
      submitExam();
    }
  }, 1000);

  restartExam();
}

// ===== TAB 2: 錯題盲測重練 =====
function renderRetryQuestions() {
  const container = document.getElementById("retryQuizContainer");
  container.innerHTML = "";

  if (wrongQuestions.length === 0) {
    container.innerHTML = `
      <div class="text-center py-16 bg-white rounded-2xl border border-slate-200">
        <span class="text-3xl block mb-2">🎉</span>
        <h3 class="text-sm font-bold text-slate-800">目前沒有累積任何錯題！</h3>
        <p class="text-xs text-slate-400 mt-1">在考場中做錯的題目會自動匯集到這裡進行盲測
重練。</p>
      </div>
    `;
    return;
  }

  wrongQuestions.forEach((q) => {
    const card = document.createElement("div");
    card.className = "bg-white border border-slate-200 rounded-xl p-5 shadow-xs";
    card.id = `retryCard-${q.id}`;

    const isMulti = q.type === "multiple" || q.answer.length > 1;
    const inputType = isMulti ? "checkbox" : "radio";

    let optionsHtml = Object.entries(q.options).map(([optKey, optVal]) => `
      <label class="flex items-start p-3 border border-slate-200 rounded-lg cursor-pointer
hover:bg-slate-50 transition">
        <input type="${inputType}" name="retry-q-${q.id}" value="${optKey}" class="mt-0.5
text-indigo-600 focus:ring-indigo-500 rounded border-slate-300">
        <span class="ml-3 text-xs leading-relaxed"><strong class="mr-1">${optKey}</strong>
${optVal}</span>
      </label>
    `);

    card.innerHTML = `
      <div class="p-4">
        <div class="text-sm font-medium text-slate-700">錯題盲測重練</div>
        <div class="text-sm text-slate-600">
          以下為您在考場中做錯的題目，請重新選擇正確答案。
        </div>
        <div class="mt-4">
          ${optionsHtml}
        </div>
      </div>
    `;
  });
}

```

```

    `).join("");

    card.innerHTML = `
      <div class="flex items-center justify-between text-xs text-slate-500 mb-2">
        <span class="bg-rose-50 text-rose-700 font-semibold px-2 py-0.5 rounded">${q.id} · 盲
測重答</span>
        <span class="font-medium text-slate-400">${q.category}</span>
      </div>

      <p class="text-sm font-medium text-slate-900 mb-4 leading-normal">${q.question}</p>
      <div class="space-y-2 mb-4">${optionsHtml}</div>

      <div class="flex items-center justify-between border-t border-slate-100 pt-3">
        <button onclick="submitRetryAnswer('${q.id}')" class="text-xs bg-rose-600
hover:bg-rose-700 text-white font-medium px-4 py-2 rounded-lg transition">
          重新驗證答案
        </button>
        <button onclick="removeWrongQuestion('${q.id}')" class="text-xs text-slate-400
hover:text-slate-600">
          移出錯題本
        </button>
      </div>

      <div id="retryResult-${q.id}" class="hidden mt-4 pt-4 border-t border-slate-100
space-y-3">
        <div id="retryBadge-${q.id}" class="text-xs font-bold inline-block px-2.5 py-1
rounded"></div>
        <div class="text-xs bg-slate-50 p-3 rounded-lg border border-slate-200 leading-relaxed
text-slate-700">
          <span class="font-bold block text-slate-900 mb-1">💡 詳解複習: </span>
          ${q.explanation}
        </div>
        <div class="flex justify-end gap-2 pt-1">
          <button onclick="switchTab('concepts')" class="text-xs text-indigo-600
hover:underline">
            📖 查看概念庫對應架構
          </button>
        </div>
      </div>
    `;

    container.appendChild(card);
  });

```

```

}

function submitRetryAnswer(qId) {
  const q = wrongQuestions.find(item => item.id === qId);
  const inputs = document.querySelectorAll(`input[name="retry-q-${qId}"]:checked`);
  const selected = Array.from(inputs).map(i => i.value).sort();

  if (selected.length === 0) {
    alert("請先選擇答案！");
    return;
  }

  const resultBox = document.getElementById(`retryResult-${qId}`);
  const badge = document.getElementById(`retryBadge-${qId}`);
  resultBox.classList.remove("hidden");

  const isCorrect = JSON.stringify(selected) === JSON.stringify(q.answer.sort());

  if (isCorrect) {
    badge.className = "text-xs font-bold inline-block px-2.5 py-1 rounded bg-emerald-100 text-emerald-800";
    badge.innerHTML = `✅ 太棒了，這次答對了！ 正解為:${q.answer.join(", ")}`;
    badge.onclick = "removeWrongQuestion(`${q.id}`)" class="ml-3 underline text-emerald-900 font-bold">點此標記為已掌握(移出)</button>`;
  } else {
    badge.className = "text-xs font-bold inline-block px-2.5 py-1 rounded bg-rose-100 text-rose-800";
    badge.innerText = `❌ 依然選錯了。正確答案為:${q.answer.join(", ")}`;
    badge.innerText += `(本次選: ${selected.join(", ")})`;
  }
}

function removeWrongQuestion(qId) {
  wrongQuestions = wrongQuestions.filter(item => item.id !== qId);
  try { localStorage.setItem("IPAS_WRONG_QUESTIONS", JSON.stringify(wrongQuestions)); }
  catch(e) {}
  updateWrongBadge();
  renderRetryQuestions();
}

// ===== TAB 3: 5 維度核心概念庫 =====
let currentConceptCat = "all";

```

```

function filterConceptCat(cat, btn) {
  document.querySelectorAll(".c-tag").forEach(b => {
    b.classList.remove("bg-indigo-600", "text-white");
    b.classList.add("bg-slate-100", "text-slate-600");
  });
  btn.classList.add("bg-indigo-600", "text-white");
  btn.classList.remove("bg-slate-100", "text-slate-600");
  currentConceptCat = cat;
  renderConcepts();
}

function filterConcepts() {
  renderConcepts();
}

function renderConcepts() {
  const container = document.getElementById("conceptListContainer");
  const keyword = (document.getElementById("conceptSearchInput")?.value ||
  "").trim().toLowerCase();
  container.innerHTML = "";

  let list = conceptDatabase.filter(c => {
    const matchCat = currentConceptCat === "all" || c.category === currentConceptCat;
    const matchText = !keyword || c.title.toLowerCase().includes(keyword) ||
    c.analysis.toLowerCase().includes(keyword) || c.glossary.toLowerCase().includes(keyword) ||
    c.paper.toLowerCase().includes(keyword);
    return matchCat && matchText;
  });

  if (list.length === 0) {
    container.innerHTML = `<div class="text-center py-12 text-slate-400 text-xs">查無符合關鍵
字的核心概念</div>`;
    return;
  }

  list.forEach(c => {
    const card = document.createElement("div");
    card.className = "bg-white border border-slate-200 rounded-2xl p-5 shadow-xs
space-y-3.5";
    card.innerHTML = `
    <div class="flex items-center justify-between border-b border-slate-100 pb-2.5">
    <div>
    <span class="text-[10px] font-bold bg-indigo-50 text-indigo-700 px-2 py-0.5

```

```

rounded">${c.category}</span>
    <h3 class="text-sm font-bold text-slate-900 mt-1">${c.title}</h3>
  </div>
  <button onclick="askConceptAI('${c.id}')" class="text-xs bg-indigo-50 text-indigo-700
hover:bg-indigo-100 px-2.5 py-1 rounded-lg font-medium transition">
    🤖 AI 深入提問
  </button>
</div>

<div class="text-xs bg-slate-50 p-3 rounded-xl border border-slate-200">
  <span class="font-bold text-slate-800 block mb-1">🎯 1. 概念分析與考查目的:</span>
  <p class="text-slate-600 leading-relaxed">${c.analysis}</p>
</div>

<div class="text-xs bg-slate-900 text-slate-200 p-3.5 rounded-xl font-mono
leading-relaxed">
  <span class="font-bold text-emerald-400 block mb-1 text-[11px]">⚙️ 2. 底層運作架構 (
Pipeline / Mechanism) :</span>
  ${c.pipeline}
</div>

<div class="text-xs bg-amber-50/60 p-3 rounded-xl border border-amber-200/60">
  <span class="font-bold text-amber-900 block mb-1">📖 3. 必考專有名詞解析 (Glossary)
: </span>
  <p class="text-amber-800 leading-relaxed">${c.glossary}</p>
</div>

<div class="text-xs bg-indigo-50/40 p-3 rounded-xl border border-indigo-100">
  <span class="font-bold text-indigo-950 block mb-1">🔧 4. 具體調校與工程手法 (
Techniques) :</span>
  <p class="text-indigo-900 whitespace-pre-wrap leading-relaxed">${c.techniques}</p>
</div>

<div class="text-xs bg-slate-100/80 p-3 rounded-xl border border-slate-200 flex
items-center justify-between">
  <div class="leading-relaxed">
    <span class="font-bold text-slate-700 block">📄 5. 經典開創論文與法規出處:</span>
    <span class="text-slate-600">${c.paper}</span>
  </div>
  ${c.paper_url ? `<a href="${c.paper_url}" target="_blank" class="text-indigo-600
hover:underline font-bold text-[11px] whitespace-nowrap ml-4">閱讀論文 ↗</a>` : ''}
</div>

```

```
<div id="conceptAiBox-${c.id}" class="hidden text-xs bg-indigo-50 p-3 rounded-xl border border-indigo-200 text-indigo-950 whitespace-pre-wrap leading-relaxed"></div>
```

```
`;  
    container.appendChild(card);  
  });  
}
```

```
async function askConceptAI(cld) {  
  const c = conceptDatabase.find(item => item.id === cld);  
  const box = document.getElementById(`conceptAiBox-${cld}`);  
  box.classList.remove("hidden");  
  box.innerText = "正在聯網向 Gemini AI 取得該概念的深度推導與學術背景...";
```

```
  
  let apiKey = null;  
  try { apiKey = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}  
  if (!apiKey) {  
    box.innerText = "請先於右上角設定 Gemini API Key ! ";  
    openKeySettings();  
    return;  
  }  
}
```

```
const prompt = `你是 iPAS AI 應用規劃師特約教授。請針對考點【${c.title}】提供深度研讀解析：  
1. 該技術在學術界最初由哪篇論文提出？解決了什麼痛點？  
2. 底層的數學概念或架構創新是什麼？  
3. 在 iPAS 考試中最常出現的 2 個單選題情境與陷阱是什麼？  
請引用論文出處，條理分明。`;
```

```
try {  
  const res = await callGemini(prompt, apiKey);  
  box.innerText = res;  
} catch (e) {  
  box.innerText = "連線失敗，請檢查 API Key。";  
}  
}
```

```
// ===== TAB 4: 記憶翻牌卡 (Flashcards) =====
```

```
function initFlashcards() {  
  flashcardList = conceptDatabase.map(c => ({  
    id: c.id,  
    category: c.category,  
    front: `【${c.title}】的核心概念與底層機制為何？`,  
    backAnswer: c.analysis,
```

```

        backMethod: c.pipeline,
        backPaper: c.paper,
        mastered: false
    }));

    document.getElementById("fcTotalCount").innerText = flashcardList.length;
    updateFlashcardMasteredCount();
}

function renderFlashcard() {
    if (flashcardList.length === 0) return;
    const c = flashcardList[currentFcIndex];
    const inner = document.getElementById("flashcardInner");
    inner.classList.remove("rotate-y-180");

    document.getElementById("fcCategory").innerText = c.category;
    document.getElementById("fcIndex").innerText = `${currentFcIndex + 1} /
    ${flashcardList.length}`;
    document.getElementById("fcFrontText").innerText = c.front;
    document.getElementById("fcBackAnswer").innerText = c.backAnswer;
    document.getElementById("fcBackMethod").innerText = c.backMethod;
    document.getElementById("fcBackPaper").innerText = c.backPaper;
}

function flipCard() {
    document.getElementById("flashcardInner").classList.toggle("rotate-y-180");
}

function nextFlashcard() {
    if (currentFcIndex < flashcardList.length - 1) {
        currentFcIndex++;
        renderFlashcard();
    } else {
        alert("已經是最後一張卡片了！");
    }
}

function prevFlashcard() {
    if (currentFcIndex > 0) {
        currentFcIndex--;
        renderFlashcard();
    }
}

```

```

function markCardMastery(isMastered) {
  if (flashcardList.length === 0) return;
  flashcardList[currentFcIndex].mastered = isMastered;
  updateFlashcardMasteredCount();
  nextFlashcard();
}

function updateFlashcardMasteredCount() {
  const count = flashcardList.filter(item => item.mastered).length;
  document.getElementById("fcMasteredCount").innerText = count;
}

function addQuestionToFlashcard(qId) {
  const q = examQuestions.find(item => item.id === qId);
  if (!q) return;

  flashcardList.push({
    id: `fc-${q.id}`,
    category: q.category,
    front: q.question,
    backAnswer: `正確答案: ${q.answer.join(", ")}\n\n${q.explanation}`,
    backMethod: "從錯題本加入之強化記憶卡片",
    backPaper: q.paper || "iPAS 歷屆考題精華",
    mastered: false
  });

  document.getElementById("fcTotalCount").innerText = flashcardList.length;
  alert('已將題目轉為記憶翻牌卡！可在【記憶翻牌卡】分頁背誦。');
}

// ===== TAB 5: 做題分析邏輯 =====
function updateAnalyticsView() {
  let tot = 0, cor = 0;
  Object.values(topicAnalytics).forEach(s => { tot += s.total; cor += s.correct; });
  const wrng = tot - cor;
  const acc = tot > 0 ? Math.round((cor / tot) * 100) : 0;

  document.getElementById("anTotal").innerText = tot;
  document.getElementById("anCorrect").innerText = cor;
  document.getElementById("anWrong").innerText = wrng;
  document.getElementById("anAccuracy").innerText = `${acc}%`;
}

```

```

const container = document.getElementById("categoryBars");
const keys = Object.keys(topicAnalytics);

if (keys.length === 0) {
  container.innerHTML = `<div class="text-slate-400 text-center py-6">尚未有作答紀錄, 請前往考場作答交卷 ! </div>`;
  return;
}

container.innerHTML = "";
keys.forEach(cat => {
  const item = topicAnalytics[cat];
  const pct = Math.round((item.correct / item.total) * 100);
  let colorClass = "bg-emerald-500";
  if (pct < 60) colorClass = "bg-rose-500";
  else if (pct < 75) colorClass = "bg-amber-500";

  const div = document.createElement("div");
  div.className = "space-y-1";
  div.innerHTML = `
    <div class="flex justify-between items-center text-[11px]">
      <span class="font-medium text-slate-700">${cat}</span>
      <span class="text-slate-500">${pct}% (${item.correct}/${item.total})</span>
    </div>
    <div class="w-full bg-slate-200 rounded-full h-2 overflow-hidden">
      <div class="${colorClass} h-2 rounded-full transition-all duration-500" style="width:
    ${pct}% "></div>
    </div>
  `;
  container.appendChild(div);
});
}

async function generateAIWeaknessReport() {
  let apiKey = null;
  try { apiKey = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}
  if (!apiKey) {
    alert("請先設定 Gemini API Key ! ");
    openKeySettings();
    return;
  }

  const reportBox = document.getElementById("aiWeaknessContent");

```

```

const btn = document.getElementById("btnGenAIReport");
btn.innerText = "⌚ 聯網診斷分析中...";
reportBox.innerText = "AI 正在交叉比對你的錯題與考科大綱，請稍候...";

const statsSummary = JSON.stringify(topicAnalytics, null, 2);
const wrongList = wrongQuestions.slice(0, 8).map(q => ` ${q.category}}
${q.question}`}).join("\n");

const prompt = `你是 iPAS AI 應用規劃師的資深總教練。以下為學員的作答分佈與錯題紀錄
：
主題掌握度：
${statsSummary}

代表性錯題：
${wrongList || '無'}

請為他量身打造「考前 7 天衝刺處方籤」：
1. 找出致命弱點領域（說明最容易失分的單元及可能混淆的原因）
2. 提煉 2 個必須熟記的經典架構或公式
3. 推薦該單元對應之學術論文或標準規範（包含年份與作者）
語氣嚴謹清晰，條理分明。`;

```

```

try {
  const res = await callGemini(prompt, apiKey);
  reportBox.innerText = res;
  btn.innerText = "✅ 診斷完成";
} catch (err) {
  reportBox.innerText = "生成失敗，請檢查 API Key 或網路狀態。";
  btn.innerText = "重新生成";
}
}

```

```

function resetAllStats() {
  if (confirm("確定要重置所有做題紀錄與錯題本嗎？")) {
    topicAnalytics = {};
    wrongQuestions = [];
    try {
      localStorage.removeItem("IPAS_TOPIC_STATS");
      localStorage.removeItem("IPAS_WRONG_QUESTIONS");
    } catch(e) {}
    initApp();
    updateAnalyticsView();
  }
}

```

```
}
```

```
// ===== AI 助教與 Gemini 整合 =====
```

```
async function askQuestionAI(qId) {  
  let apiKey = null;  
  try { apiKey = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}  
  if (!apiKey) {  
    openKeySettings();  
    return;  
  }  
}
```

```
const q = examQuestions.find(item => item.id === qId);  
const box = document.getElementById(`aiReply-${qId}`);  
box.classList.remove("hidden");  
box.innerText = "正在聯網向 Gemini AI 檢索開創論文與深入解析..."
```

```
const prompt = `你是 iPAS AI 應用規劃師考試助教。請用白話好懂的方式解說此題：  
題目: ${q.question}  
選項: ${JSON.stringify(q.options)}  
正確答案: ${q.answer.join(", ")}  
請包含：  
1. 核心觀念生活化比喻  
2. 開創性學術論文出處(作者、年份)  
3. 考前直覺判斷口訣`
```

```
try {  
  const res = await callGemini(prompt, apiKey);  
  box.innerText = res;  
} catch (e) {  
  box.innerText = "連線失敗，請檢查 API Key 或網路狀態。";  
}  
}
```

```
async function callGemini(promptText, apiKey) {  
  const url =  
`https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent?k  
ey=${apiKey}`;  
  const payload = {  
    contents: [{ parts: [{ text: promptText }] }],  
    tools: [{ googleSearch: {} }]  
  };  
  
  const res = await fetch(url, {
```

```

    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(payload)
  });
  const data = await res.json();

  if (data.candidates && data.candidates[0].content.parts[0].text) {
    let answer = data.candidates[0].content.parts[0].text;
    if (data.candidates[0].groundingMetadata &&
data.candidates[0].groundingMetadata.groundingChunks) {
      const sources = data.candidates[0].groundingMetadata.groundingChunks
        .filter(c => c.web && c.web.uri)
        .map(c => `[${c.web.title || c.web.uri}](${c.web.uri})`);
      if (sources.length > 0) {
        answer += `
即時聯網查證來源:
` + sources.slice(0, 3).join("
");
      }
    }
    return answer;
  } else {
    throw new Error("無法取得回答");
  }
}

```

```

function toggleChatDrawer() {
  let apiKey = null;
  try { apiKey = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}
  if (!apiKey) {
    openKeySettings();
    return;
  }
  document.getElementById("chatDrawer").classList.toggle("hidden");
}

```

```

async function sendChat() {
  const input = document.getElementById("chatInput");
  const text = input.value.trim();
  if (!text) return;

  let apiKey = null;
  try { apiKey = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}
  if (!apiKey) {
    openKeySettings();
    return;
  }
}

```

```

    }

    appendMessage(text, "user");
    input.value = "";
    const loadingId = appendMessage(" 正在聯網檢索學術文獻並思考中...", "ai");

    try {
        const reply = await callGemini(text, apiKey);
        document.getElementById(loadingId).innerText = reply;
    } catch (err) {
        document.getElementById(loadingId).innerText = "抱歉, 連線失敗或金鑰無效。";
    }
}

function appendMessage(text, sender) {
    const id = "msg-" + Math.random().toString(36).substr(2, 6);
    const isUser = sender === "user";
    const div = document.createElement("div");
    div.className = isUser ? "flex justify-end" : "flex justify-start";
    div.innerHTML = `
        <div id="${id}" class="max-w-[85%] p-2.5 rounded-xl whitespace-pre-wrap leading-relaxed
        ${isUser ? 'bg-indigo-600 text-white rounded-tr-none' : 'bg-white border border-slate-200
        text-slate-800 rounded-tl-none shadow-xs'}">
            ${text}
        </div>
    `;
    const box = document.getElementById("chatMessages");
    box.appendChild(div);
    box.scrollTop = box.scrollHeight;
    return id;
}

function openKeySettings() {
    document.getElementById("keyModal").classList.remove("hidden"); }
function closeKeySettings() { document.getElementById("keyModal").classList.add("hidden"); }
}

function saveKeySettings() {
    const val = document.getElementById("geminiApiKeyInput").value.trim();
    if (val) {
        try { localStorage.setItem("GEMINI_API_KEY", val); } catch(e) {}
        updateKeyStatus();
    }
    closeKeySettings();
}

```

```
}

function updateKeyStatus() {
  let key = null;
  try { key = localStorage.getItem("GEMINI_API_KEY"); } catch(e) {}
  const txt = document.getElementById("keyStatusText");
  if (key && txt) {
    txt.innerText = "已啟用";
  }
}

// 啟動身分安全檢查
checkAuthStatus();
</script>
</body>
</html>
```